Republic of the Philippines

UNIVERSITY OF SOUTHERN MINDANAO

Kabacan, Cotabato

COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY

CS 10 - Operating Systems

1st Semester 2026-2027

Title:	Laboratory Task 4
Topic:	CPU Scheduling
Name of Student:	**Rodnee Glen A. Martin**
Date Accomplished:	**September 18, 2026**

Instruction and Reminders:

- Read and answer carefully the following questions/exercises below.
- Complete the entry above with complete name and date accomplished.

At the end of this lab you should be able to:

- Explain how the FCFS (First Come, First Served) scheduling algorithm works.
- Draw a Gantt chart based on the given processes.
- Calculate Completion Time, Turnaround Time, and Waiting Time.
- Compute the Average Waiting Time of the processes.

Objective:

- Solve the following CPU scheduling problems by drawing Gantt charts and calculating Completion Time, Turnaround Time, and Waiting Time.

Process	Arrival Time	Burst Time
P1	0	5
P2	1	3
P3	2	8
P4	3	2

Task:

- Gantt Chart.
- Calculate CT, TAT, and WT for each process provide the solution table.
- Calculate the Average Waiting Time.

LABORATORY TASK 4: STUDENT OUTPUT & SOLUTIONS

Student: Rodnee Glen A. Martin **Date:** September 18, 2026

1. How First Come, First Served (FCFS) Works:

First Come, First Served (FCFS) is a non-preemptive CPU scheduling algorithm managed by a First-In, First-Out (FIFO) queue. The process with the earliest arrival time receives the CPU first and holds it until completion without preemption. Subsequent processes wait in the ready queue strictly according to arrival sequence.

2. Gantt Chart:



Execution Timeline Breakdown:

- Time 0 to 5 (5 ms): P1 arrives at t=0 and executes uninterrupted until completion at t=5.
- Time 5 to 8 (3 ms): P2 arrives at t=1, waits in queue, and executes from t=5 to t=8.
- Time 8 to 16 (8 ms): P3 arrives at t=2, waits in queue, and executes from t=8 to t=16.
- Time 16 to 18 (2 ms): P4 arrives at t=3, waits in queue, and executes from t=16 to t=18.

3. Formulas & Step-by-Step Calculations:

• Completion Time (CT) = Time when execution ends
• Turnaround Time (TAT) = CT – AT
• Waiting Time (WT) = TAT – BT

- `P1: CT = 5 | TAT = 5 – 0 = 5 ms | WT = 5 – 5 = 0 ms`
- `P2: CT = 8 | TAT = 8 – 1 = 7 ms | WT = 7 – 3 = 4 ms`
- `P3: CT = 16 | TAT = 16 – 2 = 14 ms | WT = 14 – 8 = 6 ms`
- `P4: CT = 18 | TAT = 18 – 3 = 15 ms | WT = 15 – 2 = 13 ms`

4. Solution Table:

Process	Arrival Time (AT)	Burst Time (BT)	Completion Time (CT)	Turnaround Time (TAT = CT - AT)	Waiting Time (WT = TAT - BT)
P1	0	5	5	5	0
P2	1	3	8	7	4
P3	2	8	16	14	6
P4	3	2	18	15	13
TOTAL	—	—	—	**41**	**23**
AVERAGE	—	—	—	**10.25**	**5.75**

5. Summary of Results:

• **Average Turnaround Time (ATAT):** (5 + 7 + 14 + 15) / 4 = 41 / 4 = **10.25 time units (ms)**
• **Average Waiting Time (AWT):** (0 + 4 + 6 + 13) / 4 = 23 / 4 = **5.75 time units (ms)**